

Multiparameter Case: Testing

- Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(x; \theta)$, where $\theta \in \Omega \subset \mathbb{R}^p$
- Testing:

$$H_0 : \theta \in \omega \quad \text{vs.} \quad H_1 : \theta \in \omega^c \cap \Omega$$

- For example, $X \sim N(\mu, \sigma^2)$. $\theta = (\mu, \sigma)$,
 $\Omega = \{(\mu, \sigma), -\infty < \mu < \infty, \sigma > 0\}$. Test:

$$H_0 : \mu = \mu_0, \quad \omega = \{(\mu_0, \sigma) : \sigma > 0\}$$

- $\omega \subset \Omega$ is defined in terms of q , $0 < q \leq p$, independent constraints,

$$g_1(\theta) = a_1, g_2(\theta) = a_2, \dots, g_q(\theta) = a_q$$

So that ω is a $(p - q)$ dimensional space.

- For example, $H_0 : \mu = \mu_0$. $g_1(\mu, \sigma) = \mu_0$

- Test statistic:

$$\Lambda = \frac{\max_{\theta \in \omega} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$$

- $\Lambda \leq 1$
- Large value (close to 1) of Λ suggests that H_0 is true, and small value indicates that H_1 is true.
- Reject H_0 if $\Lambda \leq c$ and

$$\alpha = \max_{\theta \in \omega} \mathbb{P}_{\theta \in \omega}(\Lambda \leq c)$$

- LRT:

$$\Lambda = \frac{L(\hat{\omega})}{L(\hat{\Omega})},$$

where $\hat{\omega}$ is the MLE in ω and $\hat{\Omega}$ is the MLE in Ω .

- Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Test: $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$.
- We performed a t -test before.
- $\Omega = \{(\mu, \sigma^2) : -\infty < \mu < \infty, \sigma^2 > 0\}$. The MLE in Ω is

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

- $\omega = \{(\mu_0, \sigma^2) : \sigma^2 > 0\}$. The MLE in ω is

$$\hat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu_0)^2$$

- Test statistic:

$$L(\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2}$$

$$L(\hat{\Omega}) = \frac{1}{(2\pi)^{n/2}} (\hat{\sigma}^2)^{-n/2} e^{-n/2}$$

$$L(\hat{\omega}) = \frac{1}{(2\pi)^{n/2}} (\hat{\sigma}_0^2)^{-n/2} e^{-n/2}$$

$$\Lambda = \left[\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \mu_0)^2} \right]^{n/2}$$

- Reject H_0 if $\Lambda \leq c$.

- Note that

$$\sum_{i=1}^n (X_i - \mu_0)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$$

- Observe that $\Lambda \leq c$ if and only if

$$\left| \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \right| \geq c'$$

Theorem. Let $\hat{\theta}_n$ be the MLE when the parameter space is Ω (a p -dimensional space) and $\hat{\theta}_{0n}$ be the MLE when the parameter space is ω (a $(p-q)$ -dimensional space). Then, under H_0 ,

$$-2 \log \Lambda \xrightarrow{D} \chi_q^2,$$

where

$$\Lambda = \frac{L(\hat{\theta}_{0n})}{L(\hat{\theta}_n)}.$$

Let $X_1, \dots, X_{n_1} \stackrel{iid}{\sim} \text{Bernoulli}(p_1)$ and $Y_1, \dots, Y_{n_2} \stackrel{iid}{\sim} \text{Bernoulli}(p_2)$.

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 \neq p_2$$

